



CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

SYLLABUS

(Choice Based Credit System)

Faculty of Technology & Engineering

Bachelor of Technology Programme

(Second Year Mechanical Engineering)

Academic Year: 2025-26



Chandubhai S Patel
Institute of Technology

Department of Mechanical Engineering – Chandubhai S Patel Institute of Technology

Vision

To be recognized as a center for outstanding education and research in field of mechanical engineering.

Mission

To produce well qualified engineers, who are innovative, contributors to their profession by catering to diverse societal needs.

Program Educational Objectives:

The objectives of the Mechanical Engineering Undergraduate Program are to produce graduate who can:

1. Apply their knowledge and expertise to find solutions of the problems in core and allied areas of engineering, exhibiting creativity and entrepreneurial skills.
2. Conduct themselves in an ethical and professional manner, exhibiting effective skills in communication, management, teamwork and leadership.
3. Continue their intellectual development through self-study and continuing education, practicing life-long learning.

Program Outcomes: Mechanical Engineering

- PO1 **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2 **Problem analysis:** Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO3 **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- PO4 **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- PO5 **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

- PO6 **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- PO7 **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- PO8 **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO9 **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- PO10 **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PO11 **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- PO12 **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes: Mechanical Engineering

- PSO1 The Mechanical Engineering graduates will be able to analyse, design, and evaluate the performance of mechanical components and systems by using various tools.
- PSO2 The Mechanical Engineering graduates will be able to plan, automate and evaluate manufacturing and system processes.



CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Bachelor of Technology (Mechanical Engineering) Programme
(Choice Based Credit System)

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
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www.charusat.ac.in

Year 2025-26

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

FACULTY OF TECHNOLOGY AND ENGINEERING ACADEMIC REGULATIONS Bachelor of Technology Programmes

Choice Based Credit System

Academic regulations recommendations are provided to ensure uniform system of education, programmes duration, eligibility criteria for admission, course credits distribution, teaching and examination pedagogy, detailed syllabus with reference material.

1. System of Education

The Charotar University of Science and Technology (CHARUSAT) shall follow Choice based Credit System (CBCS) with Semester pattern at Undergraduate and Master levels. Each semester will be at least of 90 working days. Apart from the programme core courses, provision for choosing University level electives and Programme electives are available under the CBCS.

2. Duration of Programme

Undergraduate programme	(B. Tech.)
Minimum	8 semesters (4 academic years)
Maximum	12 semesters (6 academic years)

3. Eligibility for admissions:

As enacted by the Government of Gujarat/AICTE/UGC from time to time.

4. Mode of admissions

As enacted by Govt. of Gujarat from time to time.

5. Programme structure and Credits

As per annexure – 1 attached

6. Attendance

- 6.1 Students are expected to maintain 100% attendance in all courses. However, students may involuntarily have to miss classes due to illness or some family emergency; students are permitted to maintain a minimum attendance of 75% with producing proof or reason for the absence. In case of medical exigencies, the student/parent should inform the principal immediately through call or by email. Within a week, starting from the day of absence, the proof of medical exigency must be submitted to the Principal's office.
- 6.2 Unauthorized absence will be considered as part of the discretionary 25% for fulfilling the minimum 75% attendance requirement for appearing in the examination.
- 6.3 Students nominated/sponsored by the University to represent in various forums like seminars/conferences/workshops/competitions or taking part in co-curricular/extra-curricular events will be given attendance credit provided the student applies in writing for such a leave in advance and obtains sanction from the Principal of his/her Institute for academic related requests.

7 Course Evaluation

- 7.1 The performance of every student in each course will be evaluated as follows:
 - 7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment. The respective department /institute will conduct the continuous assessment. The course faculty members shall share the pedagogy related to the continuous evaluation with the students.
 - 7.1.2 Final end-semester examination by the University through a written tablet/paper or practical test or oral test or presentation by the student or a combination of these.
 - 7.1.3 The weightages of continuous assessment and End-semester university examination in overall assessment shall depend on the individual course as approved by the Academic Council through the Board of Studies.
 - 7.1.4 The performance of a candidate in continuous assessment and in end-semester examination together (if applicable) shall be considered for deciding the final grade in a course.
 - 7.1.5 To earn the credit in a course, a student has to obtain grade other than FF.
- 7.2 Performance in continuous assessment and end-semester University Examination
 - 7.2.1 Minimum performance with respect to continuous assessment as well as end-semester university examination will be an important consideration for passing a course.

- 7.2.2 If a candidate fails to obtain minimum required overall percentage of marks (36%), student has to repeat the examination till the minimum required overall percentage obtained.

8 Grade Point System

- 8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:
- 8.2 Proposed Grading Scheme to awarding letter grade and grade point as per NEP 2020

Letter Grade	Grade Point	Grading Scheme for Mark (In %)
O (Outstanding)	10	96.0-100
A+ (Excellent)	9	86.0-95.9
A (Very Good)	8	76.0-85.9
B+ (Good)	7	66.0-75.9
B (Above Average)	6	56.0- 65.9
C (Average)	5	46.0 – 55.9
P (Pass)	4	36.0 – 45.9
F (Fail)	0	Below 36.0
Ab (Absent)	0	Absent

- The minimum passing marks for each pattern of evaluation are 36%

- 8.3 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i)
$$SGPA = \frac{\sum C_i G_i}{\sum C_i}$$
 where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
 and $i = 1$ to n , n = number of courses in the semester

- (ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
 and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

9 Award of Class

☞ The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
First Class with Distinction	$CGPA \geq 7.0 \text{ \& } \leq 10.0$
First class	$CGPA \geq 6.0 \text{ \& } < 7.0$
Second Class	$CGPA \geq 5.0 \text{ \& } < 6.0$
Pass Class	$CGPA < 5.0$

10 Detention Criteria

- ❖ A student will be promoted to next year only if he/she has cleared all the courses of the year he/she is studying in.

Link: <https://charusat.ac.in/> => Student's Corner => Detention Rules

11 Transcript:

- The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHOICE BASED CREDIT SYSTEM
FOR
BACHELOR OF TECHNOLOGY & ENGINEERING

Choice Based Credit System

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist Foundation courses, Core courses, Elective courses, Non-credit (audit) courses and SWAYAM MOOCs.

1.1 Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

1.2 Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme. Environmental science will be a compulsory University core for all Undergraduate Programmes.

B. Programme Core courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

1.3 Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialised or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline/domain or nurtures the candidates proficiency/skill is called an elective course. The following elective courses are incorporated in the CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties/programmes. As a general guideline, Programme should incorporate 2 University Electives of 2 credits each (total 4 credits).

B. Programme Elective Courses(PE):

The programme-specific pool of elective courses offered by the respective programme.

1.4 Non-Credit Course (NC) - AUDIT Course

A 'Non Credit Course' is a course where students will receive Participation or Course Completion certificate. This will be reflected in Student's Grade Sheet but the grade of the course will not be considered to calculate SGPA and CGPA. Attendance and Course Assessment is compulsory for Non Credit Courses.

1.5 Credit Transfer through SWAYAM MOOCs

CHARUSAT provides credit transfer as per UGC guidelines to all the students from SWAYAM against elective courses. The credit transfer is offered in two modes: (a) Partial credit transfer (b) Full credit transfer.

1.6 Medium of Instruction

The Medium of Instruction will be English.

In consonance with the National Education Policy (NEP) 2020 and the guidelines of the University Grants Commission (UGC), Charotar University of Science and Technology (CHARUSAT) implements the Multiple Exit scheme in their Bachelor of Technology programme.

Facilitating multiple exit options with UG certificate/UG diploma/or UG degree depending upon the number of credits secured. Skill component with progressive enhancement in skills in respective disciplines is introduced in the curriculum right from the 1st year of the program to ensure the student employability after every exit.

Students may be permitted to take a break from the study during the period of study but the total duration for completing the programme shall not exceed 7 years. If student wishes, he/she shall be allowed to re-enter the degree programme within three years and complete the degree programme within the stipulated maximum period of seven years. The exit qualifications along with credit requirements is presented in a below Table.

Table Exit Qualifications along with Credit Requirements

ACADEMIC LEVEL	EXIT QUALIFICATION AND CREDITS REQUIRED	NATIONAL CREDIT LEVEL (NCrF)
1st year of UG Degree	UG Certificate will be awarded Minimum 40 credits followed by an exit 4-credit skills enhancement course	4.5
2nd year of UG Degree	UG Diploma will be awarded Minimum of 80 credits followed by an exit 4-credit skills enhancement course	5.0
3rd year of UG Degree	B.Sc. will be awarded Minimum of 120 credits	5.5
4th year of UG Degree	B. Tech. will be awarded Minimum of 160 credits, with minimum of 40 credits each at level	6.0

Guidelines for multiple exit along with awarding UG Certificate, UG Diploma, and Degrees:

A. Exit after First Year

Award: UG Certificate in XXXX

Students must undertake a 4-credit skill enhancement course before the commencement of the 2nd year. This shall require 120 hours engagement in the relevant industry/organization. Student shall have the option to continue the 2nd year onward study.

The student shall be awarded with “UG Certificate in XXXX”, with redemption of credits from ABC. Total credits redemption shall be 1st year credits + 4 credits earned in summer.

B. Exit after Second Year

Award: UG Diploma in XXXX

Students must undertake a 4-credit summer internship/ skill enhancement course before the commencement of 3rd year. This shall require 120 hours of engagement in the relevant industry/organization. Students shall have an option to continue the 3rd year onward study.

The student shall be awarded with “UG Diploma in XXXX”, with redemption of credits from ABC. Total credits redemption shall be 1st and 2nd year credits + 4 credits earned in summer.

C. Exit after Third Year

Award: B.Sc. Degree in XXXX

The student shall be awarded with “B.Sc. Degree in XXXX”, with redemption of 3 years’ credits from ABC.

4-Credit (120 hrs.) Skills Enhancement Courses:

Level/ Branch	EE
1st year of UG Degree	MEUS102: Machine Drawing
2nd year of UG Degree	MEUS202: Summer Internship

Value Added Courses:

Inclusion of 2 credits courses on Community Service/ NSS/NCC/ Sports; and provision to earn extra credits based on undertaking Research/ Academic/Cultural/ and other Developmental activities is introduced.

- The component of ‘Value-added Courses’ could be enriched to include alternatives that could contribute to the holistic development of the students. In light of this, it is proposed to include a 02 credit Course on Community Service/ NSS/NCC/Sports as a compulsory course in the existing curricula offered across all UG programs at CHARUSAT. It is mandatory for all UG students (from 2024-25 admission batch) to undertake this course, and the course shall not account for the overall CGPA.
- Provision to earn extra credits (**Skill Augmentation** course) based on co-curricular & extracurricular activities is proposed. It was emphasized that the extra credits could help in fuller realization of the Graduate Attributes laid down by the university as well as could serve as an important ingredient facilitating the progression and recognition of fast learners. All UG students (from 2024-25 admission batch) must undertake this course during the studies and the credits of the course shall not account for the overall CGPA.

TEACHING & EXAMINATION SCHEME

S.Y B.TECH. PROGRAMME IN MECHANICAL ENGG. (Admission batch 2024-25)											
Level	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
Sem-3	HSUV201	Creativity, Problem Solving and Innovation	0	2	2	2	0	0	25	25	50
	XXXXX	University Elective - I	0	4	4	2	0	0	25	25	50
	MSUD201	Engineering Mathematics - III	4	0	4	4	50	50	0	0	100
	CLUC203	Mechanics of Solids	4	2	6	5	50	50	25	25	150
	MEUC201	Engineering Thermodynamics	3	0	3	3	50	50	0	0	100
	MEUC202	Mechanisms & Machines	4	0	4	4	50	50	0	0	100
	MEUC203	Casting & Joining Processes	3	0	3	3	50	50	0	0	100
					26	23					700
Sem-4	HSUV203	Indian Knowledge System	0	2	2	2	0	0	25	25	50
	XXXXX	University Elective - II	0	4	4	2	0	0	25	25	50
	MEUC204	Machining & Forming Processes	3	0	3	3	50	50	0	0	100
	MEUC205	Material Engineering & Metallurgy	4	2	6	5	50	50	25	25	150
	MEUS201	Manufacturing Processes Lab.	0	2	2	1	0	0	25	25	50
	MEUC206	Fluid Mechanics	3	2	5	4	50	50	25	25	150
	MEUC207	Dynamics of Machines	4	2	6	5	50	50	25	25	150
	MEUD201	Numerical Methods	0	4	4	2	0	0	25	25	50
					32	24					750

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
List of University Elective Courses
for 3rd Semester Undergraduate Programme
(Effective from Academic year 2025-26)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	MEUE201: ENGINEERING GRAPHICS AND DESIGN	ME/FTE
2	EEUE201: FUNDAMENTALS OF ELECTRICAL ENGINEERING	EE/FTE
3	CLUE201: ENVIRONMENT AND DEVELOPMENT	CL/ FTE
4	ECUE201: SCIENTIFIC COMPUTING USING MATLAB	EC/ FTE
5	CEUE201: THE JOY OF COMPUTING USING PYTHON	CE/ FTE
6	ITUE201: INTRODUCTION TO QUANTUM COMPUTING	IT/FTE
7	CSUE201: PYTHON FOR DATA SCIENCE	CSE/ FTE
8	AIUE201: AN INTRODUCTION TO ARTIFICIAL INTELLIGENCE	AIML/FTE
9	BMUD201: MONEY AND BANKING	I2IM / FMS
10	CAUD203: HUMAN COMPUTER INTERACTIONS	CMPICA /FCA
11	PTUD191: BASICS OF HEALTH PROMOTION AND EDUCATION INTERVENTION	ARIP / FMD
12	NRMD251: FIRST AID MASTERCLASS- A COMPLETE GUIDE TO FIRST AID	MTIN / FMD

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
List of University Elective Courses
for 4th Semester Undergraduate Programme
(Effective from Academic year 2025-26)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	MEUE202: NATURE AND PROPERTIES OF MATERIALS	ME/FTE
2	EEUE202: SOLAR ENERGY ENGINEERING AND TECHNOLOGY	EE/FTE
3	CLUE202: ECOLOGY AND ENVIRONMENT	CL / FTE
4	ECUE202: INTRODUCTION TO INTERNET OF THINGS	EC / FTE
5	CEUE202: SOFTWARE CONCEPTUAL DESIGN	CE / FTE
6	ITUE202: ETHICAL HACKING	IT/FTE
7	CSUE202: GOOGLE CLOUD COMPUTING FOUNDATIONS	CSE/FTE
8	AIUE202: SOCIAL NETWORK ANALYSIS	AIML/FTE
9	BMUD251: ECONOMICS OF HEALTH AND HEALTH CARE	I2IM / FMS
10	CAUD204: MODERN APPLICATION DEVELOPMENT	CMPICA / FCA
11	PTUD192: ERGONOMICS WORKPLACE ANALYSIS	ARIP / FMD
12	NRMD261: MINDFULNESS AND WELL-BEING: LIVING WITH BALANCE AND EASE	MTIN / FMD

B. Tech. (Mechanical Engineering) Programme

SYLLABI (Semester – 3)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

HSUV201: CREATIVITY, PROBLEM SOLVING AND INNOVATION
B TECH 3rd SEMESTER
Faculty of Humanities (FoH)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	02	2
Marks	--	50	--	50	

Pre-requisite courses:

Creative Problem Solving

<https://www.coursera.org/learn/creative-problem-solving>

Objectives of the Course:

To facilitate learners to:

- gain familiarity with the mechanics of creativity and problem solving.
- develop an attitude for innovation.
- develop creative thinking skills using cone of learning components leading to understanding of strategies of creativity, problem solving and innovation.
- explore applications of the concepts of creativity and problem-solving skills in personal, social, academic, and profession life.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Creativity, Problem Solving and Innovation	06
2.	Questioning, Learning and Visualization	06
3.	Creative Thinking and Problem Solving	06
4.	Logic, Language and Reasoning	06
5.	Contemporary Issues and Practices in Creativity and Problem Solving	
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

Detailed Syllabus:

1. Introduction to Creativity, Problem Solving and Innovation 06 Hours 20%
Definitions of Creativity and Innovation, Need for Problem Solving and Innovation, Scope of Creativity in various Domains, Types and Styles of Thinking, Strategies to develop Creativity, Problem Solving and Innovation

skills		
2. Questioning, Learning and Visualization	6 Hours	20%
Strategy and Methods of Questioning, Asking the Right Questions, Strategy of Learning and its Importance, Sources and Methods of Learning, Purpose and Value of Creativity Education in real life, Visualization strategies - Making thoughts Visible, Mind Mapping and Visualizing Thinking		
3. Creative Thinking and Problem Solving	6 Hours	20%
Creative Thinking and its need, Strategy of Thinking Fluency, Generating all Possibilities, SCAMPER Technique, Divergent Vs Convergent Thinking, Lateral Vs Vertical Thinking, Fusion of Ideas for Problem Solving, Applying strategies for Problem Solving		
4. Logic, Language and Reasoning	6 Hours	20%
Basic Concepts of Logic, Statement Vs Sentence, Premises Vs Conclusion, Concept of an Argument, Functions of Language: Informative, Expressive and Directive, Inductive Vs Deductive Reasoning, Critical Thinking & Creativity, Moral Reasoning		
5. Contemporary Issues and Practices in Creativity and Problem Solving	6 Hours	20%
Cognitive Research Trust Thinking for Creatively Solving Problems, Case Study on Contemporary Issues and Practices in Creativity and Problem Solving		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Demonstrate creativity in their day to day activities and academic output.
CO2	Solve personal, social and professional problems with a positive and an objective mindset.
CO3	Think creatively and work towards problem solving in a strategic way.
CO4	Initiate new and innovative practices in their chosen field of profession.
CO5	Give logical ideas, opinions, and solutions to problems.
CO6	Think critically over the situation and drawing conclusion.

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10

Total	100* (scaled to 25 marks)
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*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

Recommended Study Material:

❖ Text book:

1. R Keith Sawyer, ZigZag, The Surprising Path to Greater Creativity, Jossy-Bass Publication 2013
2. Michael Michalko, Crackling Creativity, The Secrets of Creative Genus, Ten Speed Press 2001

❖ Reference book:

1. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
2. Edward De Beno, De Beno's Thinking Course, Revised Edition, Pearson Publication 1994
3. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
4. Tony Buzan, How to Mind Map, Thorsons Publication 2002
5. Scott Berkum, The Myths of Innovation, Expanded and revised edition, Berkun Publication 2010
6. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
7. Ira Flatow, The all Laughed, Harper Publication 1992
8. Paul Sloane, Des MacHale & M.A. DiSpezio, The Ultimate Lateral & Critical Thinking Puzzle book, Sterling Publication 2002

❖ Additional Readings

1. Keith Sawyer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007
2. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
3. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002

4. Tom Kelly with Jonathan Littman, The Art of Innovation, Profile Publication 2001
5. Roger Von Oech, A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
6. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
7. Jonah Lehrer, Imagine How Creativity Works, Canongate Books Publication 2012
8. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication 1994
9. Scott G Isaksen, K Brain Doval, Donald J Treffinger, Creative Approach to Problem Solving, Sage Publication 2000
10. Donald J Treffinger, Scott G Isaksen, K Brainstead Dorval Creative Problem Solving An Introduction, Prufrock Press 2006
11. H Scott Fogler & Steven E. LeBlance, Strategies for Creative Problem Solving, Prentice Hall Publication 2008
12. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'Reilly Publication 2010.
13. Howard Gardner, Creating minds, Basic Books Publication 1993
14. Mihaly Csikszentmihalyi, Creativity—Flow and Psychology of Discovery and Invention, Harper Publication 1996
15. Martin Gerdner, W. H., Aha! Insight, Freeman Publication 1978
16. Paul Sloane, Test Your Lateral Thinking IQ, Sterling Publication 1994
17. Paul Sloane & Des MacHale Intriguing, Lateral Thinking Puzzles, Sterling Publication 1996

❖ **Web material:**

1. Internet Search based May TED talks and other sources for videos, slide shares, problems, etc

MSUD201: Engineering Mathematics III

B.TECH. 3rd SEMESTER (ME)

Credits and Hours:

Teaching Scheme	Theory		Practical		Total	Credit
Hours/week	4		-		-	4
Examination Scheme	Internal	External	Internal	External		
Marks	50	50	-	-	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Fourier Series and Its Applications	15
2	Laplace Transform and Its Applications	15
3	Vector Differential Calculus	10
4	Vector Integral Calculus	12
5	Curve Fitting	08

Total Hours (Theory): 45

B. Detailed Syllabus:

1 Fourier Series and Its Applications

15 Hours

25%

Periodic functions, Dirichlet's conditions, Trigonometric series.

Euler formula, Fourier series of periodic function

Fourier series: Discontinuous functions, Even and odd functions.

Half range Fourier series.

Solutions of heat, wave, Laplace equations by method of separation of variables and Fourier series.

2 Laplace Transform and Its Applications

15 Hours

25%

Laplace transforms as an improper integral and its existence.

Laplace transform of elementary functions and its properties.

Inverse Laplace transform and its properties.
 First and second shifting theorems. Laplace transform of derivatives and integrals.
 Convolution theorem and its application to obtain inverse Laplace transform.
 Laplace transform of periodic functions, Unit step function, Unit impulse function (Dirac delta function).
 Solving differential equations using Laplace transform.
 Applications of ODE :Mechanical vibration system, deflection of beams.

3 Vector Differential Calculus 12 Hours 20%

Revision of concepts of Vector algebra, Scalar and Vector fields.
 Gradient of a scalar functions, Directional derivatives.
 Divergence and Curl of a vector field and their properties.
 Physical interpretations of gradient, divergence and curl, Irrotational and Solenoidal vector fields.
 Scalar potential function

4 Vector Integral Calculus 12 Hours 20%

Introduction to line integrals and examples.
 Statement and examples of Green's theorem.
 Introduction to Surface and Volume integrals.
 Statements and examples of Stokes' and Gauss' divergence theorem.

5 Curve Fitting 08 Hours 20%

Normal equations.
 Fitting a straight line: $y = a + bx$ (linear regression).
 Fitting a parabola:: $y = a + bx + cx^2$
 Fitting a curve by transformation: : $y = ab^x, xy^a = b, y = ax^b$.

C. Course Outcomes (COs):

On the completion of the course one should be able to:

- CO1 Compute the problems related to the Fourier Series.
- CO2 Solve the complex problems through the concepts of Laplace transforms.
- CO3 Explain the concepts of vectors, directional derivative, gradient, vector fields, divergence and curl.
- CO4 Analyze parameterized curves and evaluate line integrals and surface integrals.
- CO5 Use simple theory about curve fitting to select the line of best fit for a data set.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	1	1	-	1	-	-	1	-	1	2	2	-
CO2	3	2	1	1	1	-	1	-	-	1	-	1	2	2	-
CO3	3	2	2	1	1	-	-	-	-	1	-	1	2	2	-
CO4	3	2	2	1	1	-	-	-	-	1	-	1	2	2	-
CO5	3	3	2	2	2	-	1	-	-	1	-	1	2	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:**Text Books:**

1. B. S. Grewal, Higher engineering mathematics, Khanna Publisher, New Delhi, 1996.
2. Debnath, Lokenath, and Dambaru Bhatta, Integral transforms and their applications, CRC press, 2014.
3. Erwin Kreyszig, Advanced Engineering Mathematics, 9th Ed., Jhon Wiley & Sons, India, 1999.

Reference Books:

1. H. K. Dass, Advanced engineering mathematics, S. Chand, 2008.
2. M. D. Greenberg, Advanced Engineering Mathematics, 2nd ed., Pearson. 1998.

Web Materials:

1. <http://mathworld.wolfram.com>
2. <http://en.wikipedia.org/wiki/Math>

CLUC203 MECHANICS OF SOLIDS
B TECH 3RD SEMESTER (CL/ME)

Credits and Hours:

Teaching Scheme	Theory		Practical		Total	Credit
Hours/week	4		2		6	5
Examination Scheme	Internal	External	Internal	External		
Marks	50	50	25	25	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	3
2	Simple Stresses and Strains	15
3	Principal Stresses and Strains	11
4	Shear Force and Bending Moment	10
5	Moment of Inertia	6
6	Bending and Shear Stresses in Homogeneous and Composite Beam Sections	10
7	Strain Energy	5

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 90

B. Detailed Syllabus:

1	Introduction	03 Hours	04%
1.1	Introduction to mechanics of deformable bodies		
1.2	Principle of superposition		
1.3	Classification of loaded bar		
1.4	Gradual, sudden, impact and shock loading		
1.5	Mechanical properties of materials		
2	Simple Stresses and Strains	15 Hours	25%
2.1	Stress and types of stress, Strain and types of strain		
2.2	Stress strain Characteristics for ductile and brittle materials		

2.3	Shear stresses and strains, Elasticity, Hook's law		
2.4	Axial and shear deformations, Axial force diagram		
2.5	Bars of varying section, Bars of uniformly varying cross section		
2.6	Analysis of stress for statically determinate structures and indeterminate structures		
2.7	Poisson's ratio, Volumetric strain, Biaxial and tri-axial deformations		
2.8	Elastic constant and relation between three elastic constants		
2.9	Stresses due to thermal effect		
3	Principal Stresses and Strains	11 Hours	18%
3.1	Introduction, Sign convention		
3.2	Transformation of stresses for a state of stresses under axial loading		
3.3	Transformation of stresses for plate under biaxial loading		
3.4	Principal plane and principal stresses		
3.5	Maximum shear stress, Element subjected to principal stresses		
3.6	Mohr's circle for stresses on an oblique section of a body subjected to direct stress in one plane and two plane with or without shear stress,		
3.7	Pure shear		
4	Shear Force and Bending Moment	10 Hours	17%
4.1	Concept of shear force and bending moment		
4.2	Sign conventions		
4.3	Relation between bending moment, shear force and rate of		
4.4	loading		
	Bending moment and shear force diagrams for statically		
4.5	determinate beams subjected to all different types of loading		
	Important points for shear force and bending moment diagram.		
5	Moment of Inertia	6 Hours	10%
5.1	Introduction, Radius of gyration		
5.2	Parallel axis theorem and perpendicular axis theorem		
5.3	Polar moment of inertia		
5.4	Moment of inertia by integration		
5.5	Moment of inertia of composite areas		

6 Bending and Shear Stresses in Homogeneous and composite beam sections 10 Hours 17%

- 6.1 Pure bending, Theory of pure bending
- 6.2 Assumptions and derivation of theory of simple bending
- 6.3 Neutral axis, moment resistance, section modulus
- 6.4 Strength of section
- 6.5 Bending stress in symmetrical, unsymmetrical and composite sections
- 6.6 Shear stresses, Shear flow
- 6.7 Shear stress distribution for various cross section

7 Strain Energy 05 Hours 09%

- 7.1 Elastic strain energy due to gradual loading, sudden loading, impact loading, shear and bending, Resilience

C. Course Outcomes (COs):

On the completion of the course one should be able to:

- CO1 Demonstrate an understanding of the basics of deformation and the relationship between stress and strain in solids.
- CO2 Analyze bending moments, shear forces, and stress distributions in statically determinate beams.
- CO3 Calculate moments of inertia for different geometrical shapes and their applications in engineering problems.
- CO4 Evaluate principal stresses, principal planes, and strain energy under various loading conditions.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3	3	3	3	3	3	2	3	3	2	2	3	3	3
CO2	3	3	3	3	3	3	3	2	3	3	2	2	3	3	3
CO3	3	3	3	3	3	2	3	2	3	2	2	2	2	2	3
	3	2	1	2	2	1	2	1	2	1	1	1	2	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

- 1) Junarkar, S.B. & Shah, H.J., Mechanics of Structures Vol-I, Charotar Publishing House
- 2) Shah, H. J., Mechanics of Solids, Charotar Publishing House
- 3) Khurmi R. S., Strength of Materials, S. Chand Publications
- 4) S. S. Bhavikatti, Strength of Materials, Vikas Publishing House Pvt. Ltd
- 5) Hibbeler, R.C., “Mechanics of Materials”, 6th SI edition

Reference Books:

- 1) Beer and Johnston, Mechanics of Materials
- 2) Gere & Timoshenko, Mechanics of Materials, CBS Publishers & Distributors, Delhi
- 3) S. Timoshenko, Strength of Materials (Part -1), D. Van Nostrand Company, Inc.
- 4) R. Subramanian, Strength of Materials, Oxford University Press
- 5) S. S. Rattan, Strength of Materials, Tata McGraw Hill Education Pvt. Ltd.
- 6) R. K. Bansal, Strength of Materials, Lakshmi Publications House Pvt. Ltd.

E Books:

- 1) Barry Dupen, Applied Strength of Materials for Engineering Technology
- 2) S. Timoshenko, Strength of Materials (Part -1), D. Van Nostrand Company, Inc

Web Materials:

- 1) https://onlinecourses.nptel.ac.in/noc19_ce18/preview#:~:text=Strength%20of%20Materials%20is%20a,basic%20behavior%20of%20such%20materials.
- 2) <https://www.youtube.com/watch?v=GkFgysZC4Vc>
- 3) <http://nptel.ac.in/syllabus/112106141/>
- 4) <http://nptel.ac.in/courses/Webcourse-contents/IIT-Delhi/Mechanics%20Of%20Solids/index.htm>

E. List of Experiments:

S. No.	List of Experiments
1	Izod Impact Test
2	Rockwell Hardness Test
3	Transverse Test on Timber
4	Compression Test on Bricks & Blocks
5	Tension Test on Various Materials

6	Tutorial: 1 Simple Stresses and Strains
7	Tutorial: 2 Shear Force and Bending Moment
8	Tutorial: 3 Moment of Inertia
9	Tutorial: 4 Principal Stresses and Strains
10	Tutorial: 5 Bending and Shear Stresses in Homogeneous and Composite Beam Sections
11	Tutorial: 6 Strain Energy

MEUC201 ENGINEERING THERMODYNAMICS

3rd Semester and 2nd Year

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Fundamental Concepts & Definitions	03
2.	First-Law of Thermodynamics	06
3.	Second Law of Thermodynamics	08
4.	Entropy	10
5.	Availability and Irreversibility	06
6.	General Thermodynamic Relationships	06
7.	Properties of Gases and Mixtures	06

Total hours (Theory): 45

Total hours (Lab): 00

Total: 45

B. Detailed Syllabus:

- 1 Fundamental Concepts & Definitions 03 hours 07%
 - 1.1 Definition and scope of Thermodynamics, microscopic and macroscopic approaches.
Thermodynamic System and Control Volume, Thermodynamic Properties, path and process, quasi-static process, Thermodynamic equilibrium, Concept of Work and Heat, Zeroth law of thermodynamics.
 - 1.2 Pure Substances: Definition of a pure substance, phase of a substance, triple point and critical points, Vapour- liquid-Solid phase in pure substance with water as example. Representation of pure substance properties on P-T and P-V diagrams, P-V-T surface, Use of mollier diagram.
- 2 First Law of Thermodynamics 06 hours 16%

- 2.1 First law of thermodynamics for a closed system undergoing (1) A cycle and (2) Change of state, Internal energy a property of the system, Perpetual motion machine of the first kind
- 2.2 Extension of the First law to control volume; Steady flow process, steady state-steady flow energy equation applied to nozzle, throttling process, Turbine & Compressor, Heat Exchanger, adiabatic mixing etc, Transient Flow Processes and its analysis, Filling and emptying process
- 3 Second Law of Thermodynamics 08 hours 18%**
- 3.1 Identifications of directions of occurrences of natural processes, Offshoot of II law from the I, Kelvin-Planck and Clausius's statement of Second law of Thermodynamic; Equivalence of the two statements, Carnot cycle analysis.
- 3.2 Reversibility and irreversibility- Conditions of reversibility with examples, Carnot Theorem. Thermodynamic temperature scale, Numericals based on above topics.
- 4 Entropy 10 hours 20%**
- 4.1 Introduction, Clausius equality and inequality, the property of entropy, Entropy change for open system and reversible processes, Change in entropy of ideal gases, Numericals.
- 4.2 Principle of increase of entropy. Third law of thermodynamics, entropy and disorder, Applications of entropy principle and Numericals.
- 5 Availability and Irreversibility 06 hours 12%**
- 5.1 Exergy concept, Available energy referred to cycle, nonflow and steady flow process, irreversibility, Second law efficiency, Numericals.
- 6 General Thermodynamic Relationships 06 hours 13%**
- 6.1 Introduction, Maxwell's equation, T-ds equations, Difference in heat capacities, Ratio of heat capacities
- 6.2 Helm-holtz and Gibbs function, Internal energy relations, Clausius-Claperyon equation, Joule-Thomson Co-efficient
- 7 Properties of Gases and Mixtures 06 hours 14%**
- 7.1 Introduction to perfect, ideal and real gases, equation of state, Van der Waal's Equation of state, Van der Waal's constants in terms of critical properties
- 7.2 law of corresponding states, compressibility factor; compressibility chart, and other equations of state
- 7.3 Gas Vapor Mixtures: Mixture of Ideal Gas, mixture of perfect gases, Dalton's Law of Partial Pressure, Amagat's Law. Concept of Variable Specific heat

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Explain the basic concepts of thermodynamics such as continuum, system, control volume, thermodynamic properties, thermodynamic equilibrium, work and heat.

CO2: Apply first law of thermodynamics to close and steady flow system.

CO3: Apply Second Law of Thermodynamics and entropy concepts to various thermodynamic systems.

CO4: Evaluate availability and irreversibility.

CO5: Derive relationship between measurable and non-measurable thermodynamic properties.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	2	1	-	-	-	-	-	-	-	-	-	1	-
CO3	3	2	-	-	-	-	-	-	-	-	-	-	1	-
CO4	3	2	1	-	-	1	-	-	-	-	-	-	1	-
CO5	3	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Recommended Study Material:

Text Books:

1. R. Yadav, “Fundamentals of Engineering Thermodynamics”, Central Pub House
2. P. K. Nag, “Engineering Thermodynamics”, Tata Mcgrraw-Hill Publications

Reference books:

1. Yunus a cengel, “Thermodynamics an engineering approach”, Tata Mc GRAW Hill
2. Moren M J, “Fundamentals of Engineering Thermodynamics”, John Wiley & Sons

3. Somasundaram S. L., “Engineering Thermodynamics”, New Age Pub.
4. Nagaraj P.B., “Basic Thermodynamics”, New Age Pub.
5. Arora C P, “Thermodynamics”, Tata Mc Graw Hill
6. Rao Y V C, “Theory and Problems of Thermodynamics”, Wiley Eastern Ltd

Reading Materials, web materials with full citations:

1. Steam Tables
2. Compressibility Chart
3. http://nptel.iitm.ac.in/courses/Webcourse-contents/IISc-BANG/Basic%20Thermodynamics/New_index1.html Web site: <http://www.wikipedia.org>

Other materials

1. International Journal of Thermal Sciences (Elsevier Publication)
2. Mechanical Engineering (IE)
3. Sadhna (Engineering Sciences) www.ios.ac.in/sadhna

MEUC202 MECHANISMS AND MACHINES

3rd Semester and 2nd Year

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	0	4	4
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Machines and Mechanism	10
2.	Synthesis and Kinematic Analysis of Mechanism	14
3.	Static and dynamic analysis of mechanism	10
4.	Cams	12
5.	Gears & gear trains	14

Total hours : 60

B. Detailed Syllabus:

- 1 Machines and Mechanism 10 Hours 15%
 - 1.1 Introduction, Machine and Mechanism, Rigid and resistant body, Kinematic pair
 - 1.2 Types of motion, Classification of Kinematic pairs, Degrees of freedom (mobility)
 - 1.3 Kinematic chain, Mechanism, Kinematic analysis of planar mechanisms
 - 1.4 Mobility analysis and range of movement, Grubler's criterion
 - 1.5 Kutzbach's criterion, Grashof's law, Inversions of Four-bar chain and slider crank mechanism
 - 1.6 Exact straight line mechanism of peaucelliers, Hart and scott- Russel, Approximate straight line mechanisms of Grasshopper, Watt and Roberts
 - 1.7 Pantograph, steering gear mechanisms, Condition for correct steering, Davis and Ackermann steering gears
 - 1.8 Linkage: Hooke's joint, double Hook's joint, dynamics of Hook's joint
 - 1.9 Oldham's coupling, Intermittent motion mechanism
- 2 Synthesis and Kinematic Analysis of Mechanism 14Hours 25%

- 2.1 Introduction, Absolute and Relative Motions, Displacement
- 2.2 Introduction, Absolute and Relative Motions, Displacement by Graphical Method (Relative velocity and acceleration)
- 2.3 Coriolis Acceleration. Acceleration analysis, Klein's Construction, Instantaneous Centre of Velocity
- 2.4 Kennedy Theorem, Angular velocity ratio theorem, Centrode: Space centrode, Body centrode
- 2.5 Dimensional Synthesis of mechanism: Definitions of Type, Number and Dimensional Synthesis, Definitions of Motion, Path and Function generation, precision position, Chebychev spacing, structural error, Freudenstein's equation, two and three position synthesis (function generation only) of four bar and slider crank mechanisms by graphical and analytical methods

3 Static and dynamic analysis of mechanism 10 Hours 15%

- 3.1 Static Force Analysis: Introduction, Basic concept of equilibrium, free body diagram, Static force analysis in various types of mechanisms by Graphical Method
- 3.2 Dynamic Force Analysis: D'Alemberts principle, inertia force, Combined static and inertia force analysis of different mechanisms, Turning – moment diagrams, fluctuation of energy, flywheels

4 Cams 12 Hours 20%

- 4.1 Definitions of cam and followers their uses, Types of Cams, Types of Followers and their motion, Cam Terminology
- 4.2 Displacement Diagrams, Motion of the Followers, Analysis of motion of followers

5 Gears & gear trains 14 Hours 25%

- 5.1 Introduction, Classification of Gears, Gear Terminology, Law of Gearing, Velocity of Sliding
- 5.2 Forms of teeth, Cycloidal Profile Teeth, Involute profile teeth, Path of Contact, Arc of the contact
- 5.3 Numbers of pairs teeth in contact, Interference in involute gears, Minimum Number of teeth
- 5.4 Interference between Rack and Pinion, Under cutting, Efficiency of Helical, Spiral, Worm, Worm Gear and Bevel Gears
- 5.5 Simple, Compound, Reverted, Epicyclic gear train, Analysis of Epicyclic gear train

C. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.

- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Identify the different mechanism and its use in machine, and calculate the degree of freedom of different mechanisms.

CO2: Synthesize the planer mechanism using vector and algebraic methods Draw different Cam profile.

CO3: Determine the velocity and acceleration of different mechanism

CO4: Analyze the action of static and dynamic forces on the mechanisms.

CO5: Draw different types cam profile.

CO6: Explain working of gear mechanism and calculate the rpm and torque for different types of gear trains.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	02	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	02	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	03	02	-	-	-	-	-	-	-	-	-	-	-	-
CO4	03	01	-	-	-	-	-	-	-	-	-	-	-	-
CO5	02	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	03	02	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Ambekar A. G, “Mechanism and Machine Theory”, Jain Brothers
2. Ratan S. S., “Theory of Machines”, Tata Mc Graw–Hill publications, New Delhi
3. Ghosh Amitabh, “Theory of Mechanisms and Machines”, East West Press

Reference books:

1. Rao J. S. and Duggipati R. V., “Mechanisms and theory Machines theory”, Wiley Eastern Ltd.
2. Shigley J. E. and Uicker J. J., “Theory of Machines and Mechanisms”, Oxford University Press.
3. Bevan Thomas, “Theory of Machines” Pearson Education India.
4. Sharma C. S. and Purohit Kamlesh, “Theory of Mechanisms and Machines”, PHI.
5. Singh Sadhu, “Theory of Machines”, 2nd Ed. Pearson Education
6. Uicker John J, “Theory of Machines and Mechanism”, Oxford University Press

Reading Materials, web materials with full citations:

1. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-Delhi/Kinematics%20of%20Machine/index.htm>
2. <http://nptel.iitm.ac.in/video.php?courseId=1018>

Programming Languages & Software’s: C, C++, MATLAB , Pro Engineer

Other materials

1. SADHNA (Engineering Science): <http://www.ias.ac.in/sadhana/>
2. Mechanical Engineering (IE)

MEUC203 CASTING AND JOINING PROCESSES

3rd Semester and 2nd Year

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Manufacturing Processes	02
2.	Melting Practices	04
3.	Metal Casting Processes	16
4.	Metal Joining Processes	19
5.	Advance Welding Processes	04

Total hours (Theory): 45

Total hours (Lab): 00

Total: 45

B. Detailed Syllabus:

1	Manufacturing Processes	02 Hours	4%
1.1	Basic Introduction, Importance of Manufacturing		
1.2	Classification and Selection of Manufacturing Processes		
2	Melting Practices	04 Hours	9%
2.1	Cupola: Charge calculations, construction		
2.2	Other Furnaces : working of induction furnace, crucible furnace, and reverberate furnace		
3	Metal Casting Processes	16 Hours	36%
3.1	Metal Casting, Sand Casting, Patterns		
3.2	Cores, Gating & Riser Systems, Casting Defects & Inspection		
3.3	Centrifugal Casting, Carbon dioxide molding process		
3.4	Investment casting, Continuous casting		
3.5	Shell molding, Die casting		

3.6 Advanced technologies in casting

4 Metal Joining Processes

19 Hours 42%

4.1 Classification of Welding, Soldering and Brazing

4.2 Gas Welding, Fuel Gases, Oxy–Acetylene Welding Equipment and techniques, Gas Cutting Processes

4.3 Types of Arc Welding, Principle of Arc, Arc Welding Equipment

4.4 ISI electrode classification, current and voltage selection for electrodes

4.5 Various arc welding processes

4.6 General principle of heat generation in resistance welding, Process details and working principle of different resistance welding processes.

5 Advanced Welding Processes

04 Hours 9%

5.1 Laser beam welding, Plasma arc welding, Electron beam welding

5.2 Thermite welding, Explosive welding, Under water welding, Electroslag welding, Friction stir welding, Flash butt welding etc.

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi–media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Choose appropriate manufacturing process for given application.

CO2: Explain different types of melting furnaces.

CO3: Explain and compare various types of casting processes in terms of their process capabilities and applications.

CO4: Explain the working of various types of welding processes.

CO5: Analyze various defects caused during casting and welding processes.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	-	-	-	-	-	-	-	-	-	-	-	1	-
CO3	2	1	-	-	1	-	-	-	-	-	-	1	-	-
CO4	2	-	-	-	-	-	-	-	2	-	-	1	1	-
CO5	2	2	-	-	-	1	-	-	-	-	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Rao P. N., “Manufacturing Technology – Foundry, Forming & Welding”, Vol. 2, 4th Edition, McGraw Hill Education (India)
2. Ghosh A. and Mallik A. K., “Manufacturing Science”, 2nd edition, East West Press-2010
3. Kalpakjian S. and Schmid S.R., “Manufacturing Engineering and Technology”, 7th Edition, Pearson-2013
4. Sharma P.C., “A Textbook of Production Technology (Manufacturing Processes)”, 8th Edition, S. Chand Publishing

Reference books:

1. Kumar Rajeev, “Manufacturing Processes”- Prentice Hall India Learning Private Limited -2014
2. Groover M. P., “Fundamentals of Modern Manufacturing”, 5th edition, Wiley-2012
3. Lindberg Roy A, "Processes and materials of manufacture", Fourth edition PHI, 1990
4. Schey J., “Introduction to Manufacturing Processes”, Tata McGraw Hill-1999
5. Parmar R.S., “Welding Processes and Technology”, Khanna Publishers
6. Jain P.L., “Principles of Foundry Technology”, 5th Edition, McGraw Hill Education (India)
7. Khanna O.P., “Foundry Technology”, Dhanpat Rai Publications
8. Khanna O.P., “Welding Technology”, Dhanpat Rai Publications
9. Ravi B., “Metal Casting: Computer Aided Design and Analysis”, PHI Publication

Reading Materials, web materials with full citations:

1. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-ROORKEE/MANUFACTURING-PROCESSES/index.htm>
2. Foundry Informatics Centre, New Delhi, <http://www.foundryinfo-india.org>

Other materials

1. Journal of Material Processing Technology (Elsevier Publication)
2. Production Engineering (IE) <http://www.ipcourseline.org>
3. Sadhna-Indian Academy of Sciences, <http://www.ias.ac.in/sadhana>
4. E-Foundry, <http://efoundry.iitb.ac.in>
5. International journal of metal casting (Springer Publication)

B. Tech. (Mechanical Engineering) Programme

SYLLABI (Semester – 4)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

MEUD201 - NUMERICAL METHODS

4th Semester and 2nd Year

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	00	04	04	2
Marks	00	50	50	

A. Outline of the Course:

Sr. No.	Title of the Unit
1.	Approximate Solutions of Nonlinear Equations and System of Linear Equations
2.	Interpolation
3.	Numerical Integration and Differentiation
4.	Numerical solution of Ordinary and Partial Differential Equation

Total hours (Lab): 60

B. Detailed Syllabus:

1 Approximate solutions of nonlinear equations and System of linear equations

- 1.1 Bisection method, False Position Method, Method of Iteration
- 1.2 Newton-Raphson method for single variable and their convergence
- 1.3 Gauss Jordan method
- 1.4 Gauss Seidel method and Gauss Jacobi method

2 Interpolation

- 2.1 Finite differences and associated operators
- 2.2 Newton's Interpolation formulas
- 2.3 Lagrange interpolation, Hermite interpolation
- 2.4 Error estimates of these formulae
- 2.5 Piecewise Interpolation : Linear Spline, Cubic Spline

3 Numerical Integration and Differentiation

- 3.1 Newton-Cotes Quadrature formula
- 3.2 Trapezoidal rule with error

- 3.3 Simpson's 1/3 rule and Simpson's 3/8 rule with respective errors
- 3.4 Romberg Integration and Gaussian Quadrature
- 3.5 Numerical Differentiation
- 4 Numerical solution of Ordinary and Partial differential equation
- 4.1 Overview of ordinary differential equations and partial differential equations
- 4.2 Euler method and Modified Euler method
- 4.3 Runge -Kutta 2nd and 4th order methods
- 4.4 Finite difference approximations to derivatives
- 4.5 Solution of Laplace Equation by Gauss Jacobi and Gauss Seidel Method
- 4.6 Solution of Heat Equation in One dimension

C. Instructional Methods and Pedagogy

- Attendance is compulsory in Laboratory
- Laboratory study involves demonstration/hands-on practice in relevance of course content.
- Experiments/Tutorials related to course will be carried out in the laboratory
- In the laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Determine approximate solution of non-linear equations and system of linear equations and examine the error using numerical methods.

CO2: Use numerical techniques for differentiation and definite integration.

CO3: Interpolate or extrapolate approximate value of dependent variable for any value of independent variable, employing only finitely many tabulated values.

CO4: Find the solution of ordinary and partial differential equation numerically.

CO5: Construct the algorithm and perform the various commands in programming language.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO01	PSO02
CO1	3	3	1	2	-	-	-	-	-	-	-	-	1	-
CO2	3	3	1	2	-	-	-	-	-	-	-	-	1	-
CO3	3	3	1	2	-	-	-	-	-	-	-	-	1	-
CO4	3	3	1	2	-	-	-	-	-	-	-	-	1	-
CO5	2	2	1	2	3	-	-	-	-	-	-	2	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Sastry, S. S. Introductory methods of numerical analysis. PHI Learning Pvt. Ltd., 2012.
2. Chapra, Steven C., and Raymond P. Canale. Numerical methods for engineers. Vol. 2. New York: McGraw-Hill, 2012.

Reference books:

1. Grewal, B. S., & Grewal, J. S. Numerical Methods in Engineering & Science: With Programs in C, C++ & MATLAB. Khanna, 2014.
2. Jain, M. K., Iyengar, S. R. K. and Jain, R. K., "Numerical Methods for Scientific and Engineering Computation", New Age Pvt. Pub, New Delhi, 2000.
3. K. E. Atkinson, Introduction to Numerical Analysis, 2nd Edn, John Wiley, 1989.

Reading Materials, web materials with full citations:

1. <http://numericalmethods.eng.usf.edu>, <http://nptel.ac.in>
2. <http://mathworld.wolfram.com/>

4th Semester and 2nd Year

Teaching Scheme	Theory	Practical	Total	Credit
Hours/Week	3	0	3	3
Marks	100	0	100	

Sr. No	Title of Unit	Minimum number of Hours
1.	Metal Cutting Lathes	08
2.	Drilling and Boring Machines	03
3.	Milling Machines	04
4.	Planers, Shapers and Slotters	03
5.	Sawing and Broaching Machines	02
6.	Grinding Machines and Abrasives	03
7.	Fundamentals of Metal Forming	07
8.	Rolling	03
9.	Forging	04
10.	Extrusion	03
11.	Drawing Operations	02
12.	Press Working and Dies	03

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- 1.4 Simple purpose lathes, Turret lathes, Horizontal and Vertical lathes.
- 2 Drilling and Boring Machines 03 Hours 07%**
- 2.1 Purpose and field of application of drilling and boring machines.
- 2.2 Upright drill processes, radial drills, Horizontal and Precision Boring Machines.
- 3 Milling Machines 04 Hours 08%**
- 3.1 Purpose and types of milling machines, general purpose milling machines.
- 3.2 Different types of milling operations.
- 3.3 Milling cutters, attachments extending the processing capabilities of general purpose milling machines.
- 4 Planers, Shapers and Slotters 03 Hours 07%**
- 4.1 Classification, Attachments extending the processing capacities of each.
- 5 Sawing and Broaching Machines 02 Hours 04%**
- 5.1 Metal sawing – classification; Reciprocating sawing machines, Circular sawing machines, Band sawing machines.
- 5.2 Types of broaching machines, Advantages and Limitations of Broaching.
- 6 Grinding Machines and Abrasives 03 Hours 07%**
- 6.1 Classifications of grinding machines, Cylindrical grinders, Internal grinders, Surface grinders, Tool and Cutter grinders.
- 6.2 Surface finishing, Abrasives, Manufacture of grinding wheels.
- 7 Fundamentals of Metal Forming Processes 07 Hours 16%**
- 7.1 Introduction, Classification of Forming Processes, Mechanics of Metal Working.
- 7.2 Various Temperatures in Metal Working, Cold and Hot Working. Formability, Strain Rate Effects on metal forming, Effects of Metallurgical Structure on Metal Forming, Hydro Static Pressure, Residual Stresses.
- 8 Rolling 03 Hours 07%**
- 8.1 Introduction and classification of Rolling processes, Principles of Metal Rolling,

Simplified Analysis of Rolling Load, Various Rolling Parameters.

8.2 Defects in rolled products and remedies of it.

9 Forging 04 Hours 08%

9.1 Introduction and classification of Forging Processes, Various Forging Operations, Forging Die Materials and Lubrication, Forge ability, Forging Defects and remedies.

10 Extrusion 03 Hours 07%

10.1 Introduction and classification of Extrusion Processes, Various Extrusion Operations, Metal Deformation and Forces in Extrusion.

10.2 Materials and Lubrication considerations in Extrusion Process, Extrusion Defects, Extrusion of Tubing, Production of Seamless Pipe and Tubing.

11 Drawing Operations 02 Hours 04%

11.1 Drawing of Rods, Wires and Tubes, Sizing.

12 Press Working and Dies 03 Hours 07%

12.1 Types of presses drive and feed mechanisms, press tools.

12.2 Various press working operations and its parameters, Elements of press, Various Metal Forming Operations.

12.3 Stock strip layout, study of sheet metal nesting software.

C. Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments/project etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Understand working of lathe, shaper and planer, drilling, milling and grinding machines.

CO2: Comprehend speed and feed mechanisms of machine tools.

CO3: Study machine tools and their operations.

CO4: Estimate machining times for machining operations on machine tools.

CO5: Gain knowledge of classification of metal forming processes, plastic deformation, flow stress, formability, strain, effect of temperature, strain rate and metallurgical structure on metal working, friction and lubrication, deformation zone geometry, workability, and residual stresses.

CO6: Understand metal forming processes like forging, drawing, rolling, extrusion, and deep drawing.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	-	-	-	-	-	-	-	1	1	-
CO2	3	1	-	-	-	-	-	-	-	-	-	1	1	-
CO3	1	2	-	1	-	-	-	-	-	-	-	3	2	2
CO4	2	3	-	1	-	-	-	-	-	-	-	1	2	1
CO5	3	1	-	1	-	-	-	-	-	-	-	1	1	-
CO6	2	2	-	-	-	-	-	-	-	-	-	1	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

Rao P. N., “Manufacturing Technology Vol. 2 – Metal Cutting and Machine Tools”, Tata McGraw Hill, 3rd Edition , 2013.

Rao P. N., “Manufacturing Technology – Foundry, Forming and Welding”, Tata McGraw Hill, 4th Edition, 2013.

Jain.R. K, “Production Technology: Manufacturing Processes, Technology and Automation”, Khanna Publishers, 17th Edition, 2011.

Reference Books:

Hajara Choudhury S. K., “Elements of Work–Shop Technology Vol. – I”, Media Promoters , 2008.

Hajara Choudhury S. K., “Elements of Work–Shop Technology Vol. –II”, Media Promoters , 2010.

Kalpajakian S., “Manufacturing, Engineering and Technology”, Pearson Education Canada, 7th Edition, 2013.

Chapman W. A., “Workshop Technology Vol. I and II and III”, CBS publishers & distributors pvt. ltd.

Web Material:

<http://nptel.iitm.ac.in/courses.php?branch=Mechanical>

Other Material:

<http://www.sciencedirect.com/science/journal/09240136>

<http://www.ieindia.org/publish/pr/pr.htm>

<http://www.ieindia.info/public.asp/me/me.htm>

MEUC205 MATERIAL ENGINEERING AND METALLURGY

4th Semester and 2nd Year

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to material Engineering and metallurgy	03
2.	Crystal Structure and Imperfection	04
3.	Phase Diagrams and Iron Carbon System	04
4.	Cast Iron and wrought iron	04
5.	Alloy Steel	04
6.	Heat Treatment of Steels	12
7.	Non-ferrous alloys	04
8.	Powder Metallurgy	06
9.	Corrosion and degradation of metals	05
10.	Materials testing and inspection	08
11.	Composite Materials	06

Total hours (Theory): 60

Total hours (Lab): 30

Total: 90

B. Detailed Syllabus:

- | | | | |
|-----|---|----------|----|
| 1 | Introduction to Material Engineering and Metallurgy | 03 Hours | 5% |
| 1.1 | Classification of Engineering Materials. | | |
| 1.2 | Engineering requirements of materials. | | |
| 1.3 | Properties of engineering materials. | | |
| 1.4 | Criteria for selection of materials for engineering applications. | | |
| 1.5 | Concept of Plastics Materials, Classification, Properties and Application of plastics. | | |
| 1.6 | Concept of Electronic Materials, Classification, Properties and Application of Electronics Materials. | | |

1.7	Types of Metallurgical Processes.		
2	Crystal Geometry and Imperfections	04 Hours	7%
2.1	Crystal systems, crystallographic points, directions and planes.		
2.2	X ray diffraction and Bragg's law.		
2.3	Imperfection in solids – point, line, and surface defects, Dislocation moments.		
3	Phase Diagrams and Iron Carbon System	04 Hours	7%
3.1	Gibbs phase rule and phase diagrams Type I, II, III.		
3.2	Iron carbon diagram.		
3.3	Structure Properties Relationship.		
4	Cast Iron and wrought iron	04 Hours	7%
4.1	Grades of C.I.		
4.2	Alloyed Cast Iron, Malleable Iron and S. G. Iron.		
4.3	Wrought Iron Properties and uses.		
5	Alloy Steel	04 Hours	7%
5.1	Classification of Steels.		
5.2	Properties and uses.		
5.3	Effects of different alloying metals.		
6	Heat Treatment of Steels	12 Hours	18%
6.1	Study of Heat-Treatment processes and TTT diagram.		
6.2	Normalizing, Annealing, Spheroidizing.		
6.3	Hardening, tempering, austempering.		
6.4	Martempering, case-hardening, nitriding, cyaniding.		
6.5	Induction hardening, flame-hardening, ageing.		
6.6	Application of above processes in mechanical components such as gears, bearing, turbine blades, crankshafts, pistons, cutting tool materials.		
7	Non-ferrous alloys	04 Hours	6%
7.1	Extraction of copper, aluminum and nickel.		
7.2	Alloys of copper, aluminum, Magnesium, lead, tin, zinc, nickel, titanium, bearing alloys.		
8	Powder Metallurgy	06 Hours	10%
8.1	Application, advantages and limitation of Powder metallurgy.		
8.2	Production of powder.		

8.3 Compacting, Sintering.

8.4 Equipment and process capability.

9 Corrosion and degradation of metals

05 Hours 7%

9.1 Meaning, causes and nature of corrosion

9.2 Intergranular corrosion (IGC), Hydrogen embrittlement

9.3 Measures of counteracting corrosion

9.4 Metal coatings, Organic coatings, Lining and cladding, Use of Corrosion inhibitors, cathodic protection against corrosion.

10 Materials testing and Inspection

08 Hours 18%

10.1 Introduction to Non-destructive testing.

10.2 Radiography Testing.

10.3 Dye Penetration Testing, Magnetic Particle Testing.

10.4 Ultrasonic Testing.

10.5 Jominy end quench test.

10.6 Macro-examination, Spark Test, Macro-etching, Fracture.

10.7 Microscopic examinations.

10.8 Electron Microscopy, Magnetic Testing.

10.9 Chemical analysis of steel and Iron.

11 Composite Materials

06 Hours 08%

11.1 Types and application of composites.

11.2 Particle reinforce composites, Influence of fiber orientation and concentration.

11.3 Fiber phase and the matrix phase, Polymer matrix composite.

11.4 Metal matrix composite, ceramic matrix composite and carbon-carbon composites.

C. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures the discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Explain the crystal structure and classification of materials and Explore various extraction processes to get the different form of materials.

CO2: Interpret the phase diagrams of materials.

CO3: Classify cast irons, wrought iron and explain their applications.

CO4: Compare different heat treatment processes to achieve desired properties of metals and alloys.

CO5: Understand the concept of materials testing and Inspection.

CO6: Develop concept of corrosion and its prevention.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	-	2	-	-	-	2	-	-	-	-	2	-	-
CO2	3	2	2	2	2	-	-	-	-	-	-	-	1	-
CO3	3	3	-	3	-	-	-	-	-	-	-	2	1	-
CO4	3	2	3	2	2	2	2	-	-	-	-	-	-	-
CO5	2	-	3	3	3	-	-	-	-	-	-	2	-	-
CO6	2	2	2	2	2	-	-	-	-	-	-	2	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Callister W. D. Jr, “Materials science and engineering : An Introduction”, Edition:-2006, Wiley India, New Delhi, India
2. Narang G. B. S. and Manchanedy K., “Materials and Metallurgy”, Khanna Pub New Delhi, India
3. Swaroop D. and Saxena M. N., “Elements of Metallurgy”, Rastogi Pub, Meerut, India
4. Dharmendra K umar and Jain S. K., “Material science and manufacturing process”, Vikas Pub House, New Delhi, India
5. Reed R. E., “Physical Metallurgy Principles”, Cengage Learning, 2009

Reference books:

1. Raghvan V., “Metallurgy for engineers”, Prentice Hall of India
2. Avner Sidney H., “Physical Metallurgy”, Tata McGraw Hill Education
3. Khanna O. P., “Material Science - A Text Book of Material Science & Metallurgy”, Dhanpat Rai Publisher

Reading Materials, web materials with full citations:

1. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIScBANG/Material%20Science/>

New_index1.html

2. <http://ocw.mit.edu/OcwWeb/web/courses/courses/index.htm#MaterialsScienceandEngineering>

Other materials

1. Metallurgical & Materials Engineering (IE)
2. Sadhna (Engineering Sciences) www.ios.ac.in/sadhna
3. NDT & E International (Elsevier Publications)

MEUC206 FLUID MECHANICS

4th Semester and 2nd Year

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Properties of fluid	02
2.	Fluid Statics	08
3.	Kinematics of fluid flow	08
4.	Fluid Dynamics	05
5.	Flow Measurement	04
6.	Vortex Flow	04
7.	Flow Through Pipes	04
8.	Viscous Flow	05
9.	Dimensional analysis	05

Total hours (Theory): 45

Total hours (Lab): 30

Total: 75

B. Detailed Syllabus:

1 Properties of fluid **02Hours 04**
%

1.1 Introduction, classification of fluids

1.2 Study of fluid properties

2 Fluid Statics **08Hours 18**
%

2.1 Pressure at a point, center of pressure on plane and curved surface

- 2.2 pressure measurement with various manometers with numerical problems
- 2.3 Buoyant force, stability of submerged body and floating body.
- 2.4 Meta centre & Meta centric height – analytical and experimental determination with problems

3 Kinematics of fluid flow **08Hours 18**
%

- 3.1 Description of fluid flow using Lagrangian and Eulerian method, Continuity equations for 2–D and 3–D flow in Cartesian coordinates of system
- 3.2 Classification of fluid flows, Stream line, path line, streak line, stream tube
- 3.3 Rotational and ir-rotational flow, circulation and vorticity, stream functions, velocity potential and potential flow, flow nets

4 Fluid Dynamics **05Hours 11**
%

- 4.1 Euler's equation along stream tube and in Cartesian Co-ordinates
- 4.2 Bernoulli's equation in one dimension flow and problems
- 4.3 Momentum equation for 2–D and 3–D flow along a stream line, Kinetic energy correction and momentum correction factor

5 Flow Measurement **04Hours 09**
%

- 5.1 Measurement of flow with pitot tube, venturimeter, orifice-meter, nozzle, bendmeter
- 5.2 Elementary theory of notches and weirs, flow in a curved path, pressure gradient and change of total energy across the streamlines

6 Vortex Flow **04Hours 09**
%

- 6.1 Introduction, Forced Vortex flow, Free Vortex Flow
- 6.2 Equation of Motion for Vortex Flow, Equation of Free & Forced Vortex Flow

7 Flow Through Pipes **04Hours 09**
%

- 7.1 Loss of Energy in pipes, Frictional loss in pipe flow, Chezy's and Darcy Equation for loss of Head Due to friction in pipes
- 7.2 Flow through compound, parallel and branched pipes

8	Viscous Flow	05Hours	11
			%

8.1 Introduction, Reynolds experiment

8.2 Flow between two parallel fixed plates, Couette flow, viscous flow through pipes, Hagen – Poiseuille's equation friction factor, Moody diagram, Loss of head due to friction

8.3 Determination of coefficient of viscosity using viscometer– Capillary tube type, rotating cylinder, falling sphere, Saybolt/ Redwood viscometer

9	Dimensional analysis	05Hours	11
			%

9.1 Introduction, Fundamental dimensions, dimensional homogeneity

9.2 Rayleigh's method and Buckingham's method

9.3 Dimensionless numbers and their significance, Hydraulic similitude, Type of models, Model testing–Model laws, Undistorted and Distorted models

C Instructional Methods and Pedagogy:

- .
- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D Course Outcomes (COs):

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On successful completion of the course, the student will be able to:

CO1: Calculate the Fluid Properties.

CO2: Solve the various fluid static as well as kinematics problems.

CO3: Apply Euler's and Bernoulli's equations for incompressible and in-viscid fluid flow.

CO4: Explain the dynamics of fluid flow and mechanics of viscous flow.

CO5: Analyze pipe flows and open-channel flows.

CO6: Perform dimensional analysis for problems in fluid mechanics.

Course Articulation Matrix:

	PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2
CO1	3	3	1	1	-	-	-	-	-	-	-	-	1	-
CO2	3	3	1	-	-	-	-	-	-	-	-	-	1	-
CO3	3	3	1	-	-	-	-	-	-	-	-	-	2	-
CO4	3	3	-	-	-	-	-	-	-	-	-	-	2	-
CO5	3	3	1	-	-	-	-	-	-	-	-	-	2	-
CO6	3	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Bansal R. K., “Fluid Mechanics”, Laxmi Publication
2. Rajput R. K., “Fluid Mechanics”, S. Chand & Co
3. Kumar D. S., “Fluid Mechanics and Fluid Power Engineering”, S K Kataria Publishers, New Delhi
4. Khurmi R. S. “Fluid Mechanics and Hydraulic Machines”, S. Chand & Co

Reference Books:

1. Subramanya K., “Theory and Applications of Fluid Mechanics”, Tata–McGraw Hill Publishing Co, 1993, New Delhi
2. White F. M. “Fluid Mechanics”, McGraw Hill, New York
3. Modi & Sheth, “Fluid mechanics & Hydraulic Machines”, Standard book house

web material:

1. <http://www.nptel.iitm.ac.in/courses/Webcourse-contents/IIT-KANPUR/FLUID-MECHANICS/ui/About-Faculty.html>

Other materials

1. Mechanical Engineering (IE)
2. Flow Measurement and Instrumentation (Elsevier Publications)
3. International Journal of Heat and Fluid Flow (Elsevier Publications)
4. Journal of Fluids and Structures (Elsevier Publications)
5. Sadhna (Engineering Sciences) www.ios.ac.in/sadhna

MEUC207 - DYNAMICS OF MACHINES

4th Semester and 2nd Year

Credits Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Mechanical vibration of single degree freedom system	12
2.	Mechanical vibration of two degree and multi-degree freedom system	10
3.	Gyroscope	10
4.	Friction	12
5.	Balancing	16

Total hours (Theory): 60

Total hours (Lab): 30

Total: 90

B. Detailed Syllabus:

- 1 Mechanical vibration of single degree freedom system 12Hours 21%
 - 1.1 Basic concepts of Vibration, Classification of vibration.
 - 1.2 Free vibration of undamped system.
 - 1.3 Free damped vibration with viscous damping, Coulomb damping.
 - 1.4 Response of an damped system under harmonic force , Response of an damped system under harmonic motion of the base and under rotating unbalance.
- 2 Mechanical vibration of two degree and multi-degree freedom system 10Hours 17%
 - 2.1 Free un-damped vibrations – Principal modes and natural frequencies, Co-ordinate coupling, vibration absorber.
 - 2.2 Multi-degree of Freedom Systems: Hozler's method, Matrix method

- | | | | |
|----------|--|----------------|------------|
| 3 | Gyroscope | 10Hours | 17% |
| 3.1 | Gyroscopic couple, gyroscopic stabilization of ship and vehicles | | |
| 3.2 | Application of the principle of gyroscopic couple to two wheelers | | |
| 3.3 | Four wheelers, planes and ships etc. | | |
|
 | | | |
| 4 | Friction | 12Hours | 21% |
| 4.1 | Introduction, Kinds of friction, Laws of friction, Coefficient of friction, Surface friction | | |
| 4.2 | Friction in screws with square thread and V threads, Pivot and collar friction, Rolling Friction | | |
|
 | | | |
| 5 | Balancing | 16Hours | 24% |
| 5.1 | Introduction, Static and dynamic balance, Balancing of rotating masses in different planes | | |
| 5.2 | Dynamics of Reciprocating Machines with Single Slider; Unbalance in Single Cylinder Engine Mechanisms | | |
| 5.3 | Unbalance in Multicylinder Engines –In–line, V–twin and Radial Engines; Balancing Techniques, Field Balancing, balancing machines. | | |

C. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi–media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory mentioned content. The Experiments of above mentioned content along with Governor and Cam dynamics will be conducted during Laboratory Hours.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Derive an equation of motion and find out natural frequency of single degree of freedom system and analyze the effect of external force in forced vibration and damped vibration.

CO2: Derive an equation of motion and find out natural frequency of two degree and multi degree of freedom system

CO3: Identify and describe the gyroscopic couple and reactive gyroscopic couple for ship, aero plane and motorcycle.

CO4: Apply theory of friction to find out the torque for the screw jack, bearing and frictional clutches.

CO5: Solve the problem of rotating and reciprocating unbalance system.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	2	-	-	-	-	-	-	-	3	-
CO2	3	3	3	2	2	-	-	-	-	-	-	-	3	-
CO3	3	2	-	1	-	-	-	-	-	-	-	-	3	-
CO4	3	3	-	-	-	-	-	-	-	-	-	-	3	-
CO5	3	2	2	2	1	-	-	-	-	-	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Ambekar A G, “Mechanism and Machine Theory”, Jain Brothers
2. Ratan S. S., “Theory of Machines”, Tata Mc Graw-Hill publications, New Delhi.
3. Ghosh Amitabha, “Theory of Mechanisms and Machines”, East West Press

Reference books:

1. Rao J. S. and Dukkupati R. V. “Mechanisms and theory Machines theory”, Wiley Eastern Ltd.
2. Shigley J. E and Uicker J. J., “Theory of Machines and Mechanisms”, Oxford University Press.
3. Norton R. L, “Kinematics & Dynamics of Machinery “,
4. Bevan Thomas, “Theory of Machines” Tata McGraw-Hill publications, New Delhi.
5. Ballaney P L, “Theory of Machine & Mechanism”
6. Singh Sadhu, “Theory of Machines”, 2nd Ed. Pearson Education
7. Uicker John J, “Theory of Machines and Mechanism”, Oxford University Press

Web Materials

1. <http://www.nptel.iitm.ac.in/>

Other materials

1. Programming Languages & software's: C, C++, MATLAB , Pro Engineer
2. SADHNA (Engineering Science): <http://www.ias.ac.in/sadhana/>

MEUS201 MANUFACTURING PROCESSES LAB

4th Semester and 2nd Year

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	1
Marks	0	50	50	

A. Outline of the Course:

The lab practice consists of the experiments as decided by the course Supervisors of the program core courses namely Casting and Joining processes, Machining and Forming Processes.

Sr. No.	Title	Minimum number of hours
1.	Mold making in sand casting	02
2.	Measurement of molding sand properties	02
3.	Melting and Pouring demonstration	02
4.	Gas Welding and Gas Cutting	02
5.	MIG & TIG Welding Process	02
6.	Resistance welding & Friction welding	02
7.	Dye Penetrant (DP) Test for weld defects	02
8.	Lathe Machine : Turning and Thread Cutting	06
9.	Capstan and Turret Lathe	02
10.	Drilling Machine & Spur Gear Cutting on Milling Machine	02
11.	Shaping and Planing Machine & Grinding Machine	02
12.	Different metal forming processes: Rolling Operation Using Three Roller Bending Machine, Bending Operation Using Hydraulic Pipe Bender, Press and Press Working Operations	02
13.	Forging Operation	02

Total hours (Lab): 30 hours

B. Instructional Method and Pedagogy:

- Attendance is compulsory in Laboratory
- Laboratory study involves demonstration/hands-on practice in relevance of course content.
- Experiments related to course will be carried out in the laboratory
- In the laboratory discipline and behavior will be observed strictly.

C. Course Outcomes (Cos)

At the end of the course, the students will be able to

CO1: **Prepare** the sand mold for the given pattern using sand casting process.

CO2: **Identify** the specific welding process for given application and **use** the same.

CO3: **Perform** various machining operations on different machine tools and **prepare** a job on lathe machine using given engineering drawing.

CO4: **Perform** metal forming processes like forging, drawing, rolling, extrusion, and deep drawing.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	1	1	2
CO2	1	1	2	1
CO3	2	1	1	2
CO4	1	1	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Rao P. N., “Manufacturing Technology – Foundry, Forming & Welding”, Vol. 2, 4th Edition, McGraw Hill Education (India)
2. Chapman W. A., “Workshop Technology Vol. I and II and III”, CBS publishers & distributors pvt. Ltd.

Reference books:

1. Khanna O.P., “Foundry Technology”, Dhanpat Rai Publications
2. Khanna O.P., “Welding Technology”, Dhanpat Rai Publications
3. Parmar R.S., “Welding Processes and Technology”, Khanna Publishers

4. Ravi B., “Metal Casting: Computer Aided Design and Analysis”, PHI Publication
5. Hajara Choudhury S. K., “Elements of Work–Shop Technology Vol. – I”, Media Promoters , 2008.
6. Hajara Choudhury S. K., “Elements of Work–Shop Technology Vol. –II”, Media Promoters , 2010.
7. Kalpakajian S., “Manufacturing, Engineering and Technology”, Pearson Education Canada, 7th Edition, 2013.

Reading Materials, web materials with full citations:

1. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-ROORKEE/MANUFACTURING-PROCESSES/index.htm>
2. Foundry Informatics Centre, New Delhi, <http://www.foundryinfo-india.org>

Other materials

1. Journal of Material Processing Technology (Elsevier Publication)
2. Production Engineering (IE) <http://www.ipcourseline.org>
3. Sadhana-Indian Academy of Sciences, <http://www.ias.ac.in/sadhana>
4. E-Foundry, <http://efoundry.iitb.ac.in>
5. International journal of metal casting (Springer Publication)

HSUV203: Indian Knowledge System
4th Semester and 2nd Year

Faculty of Humanities (FoH)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	02	02
Marks	--	50	--	50	

Objectives of the Course:

- to develop a foundational understanding of the Indian Knowledge System, its origins, and its significance in shaping Indian culture and society.
- to analyze the Vedic texts, their content, and their cultural and philosophical implications.
- to investigate major philosophical schools in India, including their core tenets and historical context.
- to understand the continuity and evolution of knowledge and wisdom in India across different time periods.
- to gain insight into the role of linguistics and languages in preserving and transmitting knowledge within the Indian context.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	An overview of Indian Knowledge System (IKS):	04
2.	The Vedic Corpus:	06
3.	Indian Philosophical Systems:	04
4.	Wisdom through the Ages:	04
5.	Linguistics:	06
6.	The knowledge triangle:	06
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

Detailed Syllabus:

- 1 **An overview of Indian Knowledge System (IKS):** 04
- Importance of Ancient Knowledge
 - Definition of IKS

▪ Classification framework of IKS	
▪ Unique aspects of IKS	
2 The Vedic Corpus:	06
▪ Vedas and Vedangas	
▪ Distinctive features of Vedic Life	
3 Indian Philosophical Systems:	04
▪ Introduction to Indian Philosophies	
▪ Different schools of philosophy	
4 Wisdom through the Ages:	04
▪ Puranas	
▪ Ithihasas	
▪ Niti shastras	
▪ Subhasitas	
5 Linguistics:	06
▪ Components of a language	
▪ Paṇini's work on Sanskrit grammar	
▪ Phonetics in Sanskrit	
▪ The role of Sanskrit in natural language processing	
6 The knowledge triangle:	06
▪ Prameya	
▪ Pramaṇa	
▪ Saṃsaya	
▪ Framework for establishing Valid Knowledge	
▪ Potential fallacies in the reasoning process	

Course Outcome (COs):

At the end of the course, the students will be able to

- CO1 describe the fundamental concepts and components of the Indian Knowledge System, including its historical development.
- CO2 analyze select Vedic texts, extracting their philosophical and cultural significance.
- CO3 differentiate between major Indian philosophical systems, such as Vedanta, Nyaya, and Sankhya, and comprehend their key principles.
- CO4 trace the evolution of wisdom and knowledge in India from ancient times to the present day, recognizing its continuity.
- CO5 examine the role of language and linguistics in preserving and transmitting knowledge and cultural heritage.
- CO6 discuss the interconnectedness of philosophy, science, and spirituality in IKS, highlighting their contributions to holistic understanding.

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

Recommended Study Material:

❖ Text book:

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❖ Reference book:

9. "Indian Knowledge Systems" by Kapil Kapoor
10. "Indian Knowledge Tradition" by K. Ramasubramanian
11. "The Advaita Worldview: God, World, and Humanity" by Anantanand Rambachan
12. "The Vedas: An Introduction to the Sacred Texts of India" by Roshen Dalal
13. "The Rigveda: 3-Volume Set" translated by Stephanie W. Jamison and Joel P. Brereton
14. "The Yajur Veda: A Rendering of the Jewel of the Vedic Hymns" translated by Devi Chand
15. "A History of Indian Philosophy" by Surendranath Dasgupta (Multi-volume work)
16. "Indian Philosophy and Modern Culture" by Satyendra Nath Dasgupta
17. "The Wisdom of the Upanishads" by A. Parthasarathy
18. "The Dhammapada: The Sayings of the Buddha" translated by Eknath Easwaran
19. "Prameya Mala" by Swami Tattvavidananda Saraswati
20. "Pramana: Vaisheshika Sutra of Kanada" translated by Satish Chandra Vidyabhusana
21. "Samsāya in Indian Philosophy: Text and Commentaries" by Bibhuti Baruah
22. "The Art of Reasoning" by David Kelley

MEUS202 Summer Internship 4th Semester and 2nd Year

Credit Hours:

Teaching Scheme	Total	Credit
Hours	120	4
Marks	100	

A. Outline of the Course:

- Summer internship shall be at least 4 weeks (120 hours) during the summer vacation only.
- During the Summer Internship students should enhance the skill like mechanical drafting and manufacturing technology etc.
- Department/Institute will help students to find an appropriate company/industry/organization for the summer internship.
- The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline.
- Students shall commence the internship after the approval of the department Coordinator. Summer internships in research center is also allowed.
- During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and also those of the University.
- Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.
- The student shall submit two documents to the Coordinator for the evaluation of the summer internship: (1) Summer Internship Report (2) Summer Internship Assessment Form
- Student should submit the hardcopy of summer internship report to the Coordinator.
- The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for

revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.

B. Format of Summer Internship Report

The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers) DESCRIPTION OF THE COMPANY/INDUSTRY/ORGANISATION: Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding location and spread of the company Number of employees, engineers, technicians, administrators in the company Divisions of the company, Your group and division, Administrative tree (if available) Main functions of the company, Customer profile and market share.

INTRODUCTION: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.

PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.

SOLUTION: In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.

CONCLUSIONS: In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.

REFERENCES: List any source you have used in the document including books, articles and web sites in a consistent format.

APPENDICES: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number.

C. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Integrate class room theory with real-world engineering problems of industry/organization

CO2: Develop work habits, attitudes and ethics necessary for industrial job.

CO3: Develop communication, interpersonal and other critical skills which can help them during job.

CO4: Develop professional report writing skill and presentation of their work in industry as well as at institute.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2	-	1	-	-	-	-	-	3	-
CO2	-	-	-	-	-	3	-	3	3	2	-	-	-	-
CO3	-	-	-	-	-	2	-	-	1	3	2	2	-	-
CO4	-	-	-	-	1	-	-	-	1	3	2	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No cor

MEUE203: BASICS OF ELECTRICAL VEHICLES

4th Semester and 2nd Year

(Minor Specialization Subject – I)

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	00	100	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Electric Vehicles	14
2.	Automotive Design	14
3.	Design of Electric and Hybrid Electric Vehicles	15
4.	Regulation and Standardization of Vehicles	02

Total Hours (Theory): 45

Total Hours: 45

C. Detailed Syllabus:

- | | | | |
|-----|--|---------|--------|
| 1 | Introduction to Electric Vehicles | 14Hours | 31.11% |
| 1.1 | I.C Engine powered Vehicles Vs Electrical Vehicles (EVs), How EV works, Types of EVs, Hybrid Electric Drive-train | | |
| 1.2 | Tractive effort in normal driving, Energy consumption Concept of Hybrid Electric Drive Trains, Architecture of Hybrid Electric Drive Trains, Series Hybrid Electric Drive Trains, Parallel hybrid electric drive trains, Electric Propulsion unit. | | |
| 2 | Automotive Design | 14Hours | 31.11% |
| 2.1 | Major parts of automobiles, Chassis, Frame, Body of automobile, Transmission system, Steering and Breaking systems, Suspension, Accessories. | | |
| 3 | Design of Electric and Hybrid Electric Vehicles | 15Hours | 33.34% |

- 3.1 Series Hybrid Electric Drive Train Design: Operating patterns, control strategies, Sizing of major components, power rating of traction motor, power rating of engine/generator, design of PPS
- 3.2 Parallel Hybrid Electric Drive Train Design: Control strategies of parallel hybrid drive train, design of engine power capacity, design of electric motor drive capacity, transmission design, energy storage design

4 Regulation and Standardization of Vehicles

02Hours 4.44%

- 4.1 Motor vehicle act, registration of motor vehicles, driving license, control of traffic, insurance against third party, claims for compensation, traffic signs, central motor vehicle rules, vehicle safety standards and regulations, classification and definition of vehicles, enforcement of emission norms, duties of surveyor.

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 5 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Industrial Visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of assignment.

E. **Students Learning Outcomes:** At the end of the course, the students will be able to:

CO1: Differentiate between the Electrical Vehicles and conventional vehicles.

- CO2:** Explain the needs and working of various systems in automobiles.
- CO3:** Analyze different power converter topology used for electric vehicle application.
- CO4:** Realize the Regulation and standardization of vehicles.

F. Recommended Study Material:

Text Books:

1. Singh Kirpal, "Automobile Engineering" Volume I & II, Standard Pub.& Dist.
2. M. Ehsani, Y. Gao, S. Gay and Ali Emadi, Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory, and Design, CRC Press, 2005.
3. Iqbal Husain, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press, 2003.
4. Gupta K. M. "Automobile Engineering" Volume I & II, Umesh Pub.
5. Gupta R. B. "Automobile Engineering", Satya Prakashan.
6. Giri N. K., "Automobile Technology", Khanna Pub.

Reference Books:

1. Crouse W., "Automotive Mechanics", Tata Mc Graw Hill.
2. Narang G. B. S. "Automobile Engineering", Khanna Pub.
3. John Lowry and James Larminie, "Electric Vehicle Technology Explained", Wiley

Web Material:

1. <http://nptel.iitm.ac.in>
2. <http://www.sae.org/pubs/automotive/>
3. <https://www.springer.com/journal/12239/>

Other Material:

1. International Journal of Automotive Technology (www.ijat.net/)
2. International Journal of Automotive Technology

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING

MEUE201- ENGINEERING GRAPHICS AND DESIGN
B TECH 3rd SEMESTER (UNIVERSITY ELECTIVE-I)

Credits Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2		2	2
Marks	00	50	50	

A. Objective of the Course:

- To introduce the student to the universal language and tool of communication of engineers.
- To make them thorough in understanding and using the various concepts of Engineering Drawing.

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1	Introduction to Engineering Drawing	05
2.	Visualization of Objects	10
3.	Sectional View of Objects	08
4.	Assembly and Detailed Drawing	04
5.	Introduction to Computer Aided Drawing	03

Total hours : 30

C. Detailed Syllabus:

- | | | | |
|-----|---|----------|-----|
| 1 | Introduction to Engineering Drawing | 05 Hours | 17% |
| 1.1 | Basic of engineering graphics. | | |
| 1.2 | Lines, types of lines, use of different line in engineering drawing. | | |
| 1.3 | Dimensioning, placing of dimensions, general rules of dimensions. | | |
| 2 | Visualization of Objects | 10 Hours | 34% |
| 2.1 | Principle of visualization of objects, need of visualization, methods of visualization of three | | |

dimensional objects.

- 2.2 Interpretation of line and area in orthographic drawing.
- 2.3 Visualization of objects -pictorial view and orthographic view.
- 2.4 Visualization of objects- Isometric view.

3 Sectional view of Objects 08 Hours 27%

- 3.1 Principle of sectional view of objects,
- 3.2 Classification of sectional views,
- 3.3 Full sectional views, half sectional views, convention for sectioning.

4 Assembly and Detailed Drawing 04 Hours 12%

- 4.1 Assembly drawing with sectioning from given detailed
- 4.2 Detail drawing from given assembly

5. Introduction to Computer Aided Drawing 03 Hours 10%

- 5.1 Introduction to 2D Drawing facilities in CAD software
- 5.2 Introduction to 3D modeling & its relationship with 2D drawing views

D. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

E. Student Learning Outcomes / objectives:

Upon successful completion of this course, the student will be able to:

- 1. Understand the basics of drawing which is used in industries.
- 2. To convert sketches to engineered drawings will increase.
- 3. Improve their visualization skills so that they can apply these skills in developing new products.

4. Know the fundamental of Computer Aided Drawing & 3D Modeling.

F. Recommended Study Material:

Text Books:

1. Shah, P. J., Engineering Drawing Vol. I & II, S. Chand & Co.
2. Bhatt, N. D., Engineering Drawing, Charotar Publishing House

Reference books:

8. Gopal Krishna K.L., Engineering Drawing, Subhas Publications
9. Venugopal, K., Engineering Drawing made Easy, Wiley Eastern Ltd.
10. Agrawal, M.L. & Garg, R.K., Engineering Drawing Vol-I, Dhanpatrai & Co.
11. French, T.E., Vierck, C.J. & Foster, R. J., Graphic Science and Design, McGraw Hill
12. Luzadder, W. J. & Duff, J.M., Fundamentals of Engg. Drawing, Prentice Hall
13. Venugopal, K., Engg. Drawing and Graphics, New Age international Pry. Ltd.

NPTEL Web Link: <https://archive.nptel.ac.in/courses/112/102/112102304/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING DEPARTMENT OF
MECHANICAL ENGINEERING

MEUE202 NATURE AND PROPERTIES OF MATERIALS
B TECH 4th SEMESTER (UNIVERSITY ELECTIVE-II)

Credits Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2		2	2
Marks	00	50	50	

A. Objective of the Course:

- To give an overview on engineering requirements of materials and the importance of structure property relations of engineering materials on their selections and use.
- To understand the testing/ evaluation of strength and other properties of engineering materials.

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1	Introduction to Material Science	05
2.	Metals and Alloys	09
3.	Non-Metallic Materials	08
4.	Materials Testing and Inspection	08

Total hours : 30

C. Detailed Syllabus:

1	Introduction to Material Science	05 Hours	16%
1.1	Classification of engineering materials		
1.2	Engineering requirements of materials		
1.3	Properties of engineering materials		
1.4	Criteria for selection of materials for engineering applications		
1.5	Concept of electronic materials, classification, properties and application of electronics materials		
2	Metals & Alloys	09 Hours	34%
2.1	Classification of Metals		

- 2.2 Ferrous Metals: Classification, Composition, Properties, Application (for plain carbon steel, alloy steel including stainless steel and cast iron)
- 2.3 Non-Ferrous Metals: Classification, Composition, Properties, Application (for copper, copper alloys, aluminum and aluminum alloys)
-
- 3 Non-Metallic Materials 08 Hours 25%**
- 3.1 Introduction and classification of non-metallic materials
- 3.2 Classification of polymers on basis of thermal behavior (thermoplastics & thermosetting), properties and applications of polymers
- 3.3 Composites: Introduction, Characteristics of composites, Types and applications of composites
- 3.4 Other non-metallic materials-types, properties and applications. (like plastic, rubber, ceramics, refractories, insulators, wood etc.)
-
- 4. Materials testing and inspection 08 Hours 25%**
- 4.1 Introduction to Non-destructive testing
- 4.2 Radiography Testing
- 4.3 Dye Penetration Testing, Magnetic Particle Testing
- 4.4 Ultrasonic Testing
- 4.5 Macro-examination, Spark Test, Macro-etching, Fracture, Microscopic examinations

D. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

E. Student Learning Outcomes:

Upon successful completion of this course, the student will be able to:

1. Understand basic engineering properties of materials and its Application.
2. Understand fundamentals of various engineering steels and alloys and should have ability to select suitable materials based on their properties.
3. Classify non-metallic materials.
4. Examine various engineering materials.

F Recommended Study Material:**Text Books**

1. Narang G. B. S. and Manchanedy K., "Materials and Metallurgy", Khanna Pub New Delhi, India
2. Dharmendra K umar and Jain S. K., "Material science and manufacturing process", Vikas Pub House, New Delhi, India
3. R.S.Khurmi and R.S.Sedha, "Material Science", S.Chand
4. D.S.Nutt, "Materials science and metallurgy", S.K.Katariya and sons, Delhi

Reference Book:

1. Raghvan V., "Metallurgy for engineers", Prentice Hall of India
2. Khanna O. P., "Material Science - A Text Book of Material Science & Metallurgy", DhanpatRai Pub

NPTEL Web Link: <https://nptel.ac.in/courses/112104203>